



VisionSense Pro

Smart Disposal & Real-Time AI Object Detection System

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Problem Statement

Improper waste disposal and lack of awareness about recycling and sustainability practices remain major global challenges. Many individuals are unaware of how to correctly dispose of different types of waste, leading to environmental pollution and inefficient waste management systems. Additionally, visually impaired individuals face difficulty in identifying everyday objects. There is a need for a smart, accessible, and real-time solution that can identify objects and provide guidance for eco-friendly disposal.

Proposed Solution

VisionSense Pro is a web-based AI-powered application that performs real-time object detection using a device's webcam. The system identifies objects instantly and provides audio feedback along with eco-friendly disposal instructions. It runs entirely in the browser without requiring any installation or backend server, making it highly accessible and efficient.

Methodology / Working

1. AI Detection Engine: The system uses Hugging Face Transformers.js library with two models: OWL-ViT for open vocabulary detection and DETR-ResNet-50 for faster fallback detection. 2. Camera Processing Pipeline: The application captures frames from the webcam at regular intervals and processes them through the AI model. 3. Voice Assistance: Using the Web Speech API, the system converts detected object names into speech. 4. Smart Disposal Guidance: The system categorizes objects into recycle, trash, hazardous, compost, or donate and provides disposal steps.

Expected Impact

- Promotes proper waste segregation and environmental awareness.
- Helps visually impaired individuals identify objects using audio feedback.
- Supports Sustainable Development Goals (SDGs) such as Climate Action and Sustainable Cities.
- Provides an accessible AI solution without expensive hardware.

Project Interface

